ROLL NO: _____

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

END-SEM EXAMINATION — DECEMBER / JUNE 2024

FIRST YEAR (COMMON TO ALL BRANCHES) — SEMESTER 1

Time Allowed: 3 Hours **Maximum Marks: 70**

NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.
2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].
3. Assume suitable missing data, if any, and state it clearly.
4. Use of scientific non-programmable calculator is permitted.

--- UNIT I ---

Q.1 Explain thin film interference by reflected light. Describe Newton's Rings experiment. [14 Marks]

[OR]

Q.1 (Alt) Compare Fresnel and Fraunhofer diffraction. Derive single-slit diffraction intensity. [14 Marks]

--- UNIT II ---

Q.2 Explain Einstein's A and B coefficients. Describe construction of He-Ne and Ruby lasers. [14 Marks]

[OR]

Q.2 (Alt) Derive Numerical Aperture (NA) and Acceptance Angle of optical fiber. Step vs Graded index. [14 Marks]

--- UNIT III ---

Q.3 Explain de Broglie hypothesis and Davisson-Germer experiment. State Heisenberg Uncertainty. [14 Marks]

[OR]

Q.3 (Alt) Derive time-independent Schrodinger equation and calculate 1D infinite potential well energies. [14 Marks]

--- UNIT IV ---

Q.4 Explain Miller Indices and Bragg's Law of X-ray diffraction. [14 Marks]

[OR]

Q.4 (Alt) Explain Superconductivity and Meissner Effect. Differentiate Type-I vs Type-II superconductors. [14 Marks]

--- UNIT V ---

Q.5 State Maxwell's four electromagnetic equations. State and prove Poynting's Theorem. [14 Marks]

[OR]

Q.5 (Alt) Explain Quantum Confinement (Wells, Wires, Dots) and Sol-Gel synthesis of nanomaterials. [14 Marks]

**** END OF QUESTION PAPER — MITS GWALIOR AUTONOMOUS EXAMINATION CELL ****